1. INTRODUCTION

This document presents a detailed technical analysis and step-by-step methodology for solving the assigned cybersecurity challenges as part of the Security Intern task round.

The objective of this assessment is to demonstrate practical skills in:

1. [OSINT] Social Media
2. [Reverse] The Vault
3. [Forensics] It’s Stream-Ing
4. [Reverse] Beowulf

OSINT Report

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Scope & Objective

OBJECTIVE: This OSINT investigation report analyses the social media account 'emp_exch' to identify its online presence, linked data and behavioural patterns across various social media/public platforms.

NOTE: All information gathered is from publicly accessible sources. No attack/exploit/phishing was conducted.

Methodology

- Search the username across social media platforms.
- Collect available profile information.
- Cross-check details like emails, photos, and links.
- Review posts and activity history for patterns.
- Create a basic identity timeline.
- Document findings with screenshots.

Tools Used

1. Google Dorks
2. Sherlock
3. whoami (Chrome extension)
4. Namechk
5. whatsmyname
6. whatsmyipaddress(Email analysis, IP lookup & domain-related information)
7. whois

Investigation Steps

1. Identify Platform Presence

A. Google Dorks

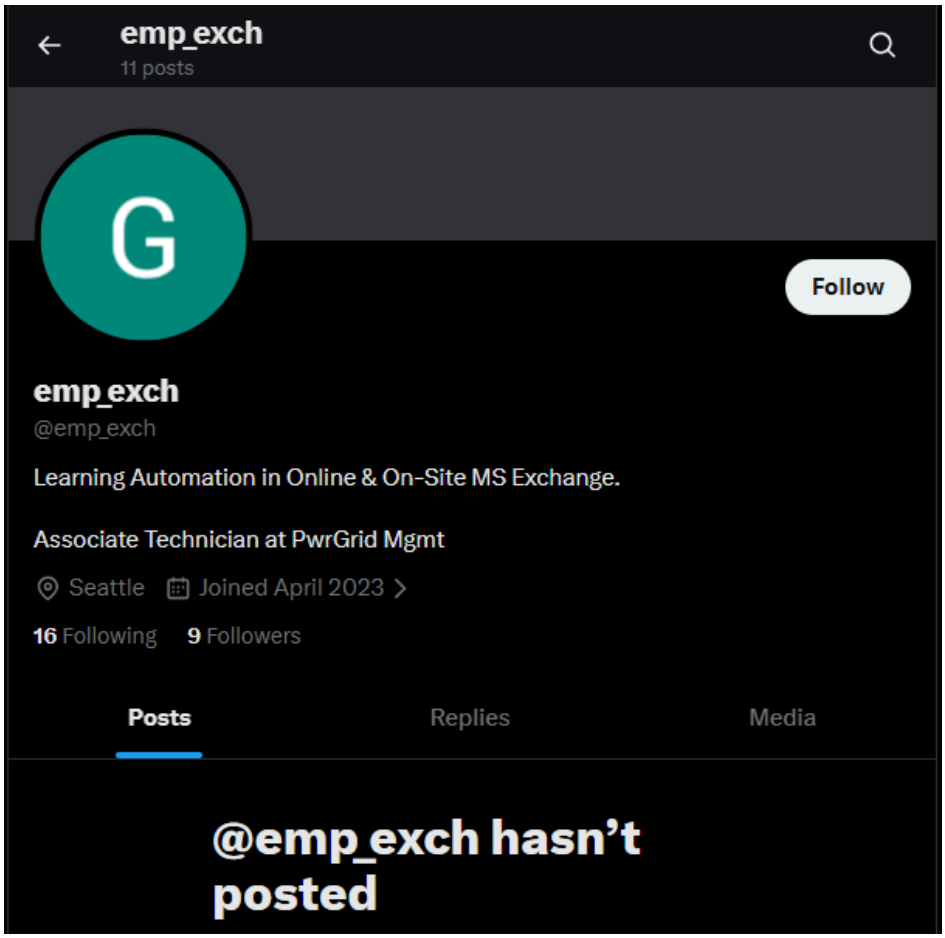
"emp_exch"

intitle: emp_exch

"emp_exch" filetype:pdf

"emp_exch" filetype:txt

Findings: The username emp_exch was found on X and in multiple Scribd documents, mainly related to DevOps and OSINT, showing an active online presence.



B. Namechk

Visited https://namechk.com and checked “emp_exch” user name.

Findings: The username is already on Facebook, Twitter, LinkedIn, Instagram, Tumblr, Pinterest, YouTube, and Flickr. Only the Twitter (X) account is active; all other platforms return 404 and are not live.

Social Media Usernames learn more

facebook	already taken	View
twitter	already taken	View
linkedin	already taken	View
instagram	already taken	View
tumblr	already taken	View
pinterest	already taken	View
youtube	already taken	View
flickr	already taken	View

C. Whatsmyname

Visited https://whatsmyname.me and checked “emp_exch” user name.

Findings: Only the X (Twitter) account is active; others show errors or link to CTF/OSINT reports, not real profiles.

whatsmyname.me/#social-results-container

emp_exch

Search

Social Media | Search Results | Email Leak Checker

X 100% social
https://x.com/emp_exch
Exists: Status 200 and contains ""reason":"taken"" (HTTP 200)

Arch Linux GitLab 100% social
https://gitlab.archlinux.org/emp_exch
Exists: Status 200 and contains ""id":" (HTTP 200)

Sourceforge 100% coding
https://sourceforge.net/u/emp_exch/profile
Exists: Status 400 and contains ""error": "invalid"" (HTTP 400)

Udemy 100% tech
https://www.udemy.com/user/emp_exch/
Exists: Status 200 and contains "| Udemy" (HTTP 200)

Znanija 100% misc
https://znanija.com/graphql/ru?operationName=Nic
Exists: Status 200 and contains ""isAvailable":false" (HTTP 200)

What is my Name | AI Assistant | English

About 3 results (0.40s) Query: "emp_exch" OR "@emp_exch" | Open in Google →

1. WWW.SCRIBD.COM
OSINT Investigation Report On Social Media Account - Emp - Exch
OSINT Investigation Report on Social Media Account_ emp_exch - Free download as PDF File (.pdf), Text File (.txt) or read online for free.

2. X.COM
My DevOps Progress v.1 :
emp_exch profile. emp_exch. emp_exch. Apr 12. My DevOps Progress v.1 : bit.ly/3Ktqbvq · 0. 0. 2. 840 · · Explore Trending StoriesGo to HomeSearch XNews.

3. X.COM
emp_exch (@emp_exch) / Posts / X - Twitter
emp_exch (@emp_exch) - Posts - Learning Automation in Online & On-Site MS Exchange. Associate Technician at PwrGrid Mgmt | X (formerly Twitter)

Learn

What is my Name | AI Assistant

STATUS
All | Found | Not Found

CATEGORY
socia
All Categories | Social network

Show NSFW content

Pinterest Found
https://www.pinterest.com/emp_exch/
Social Network | 312ms

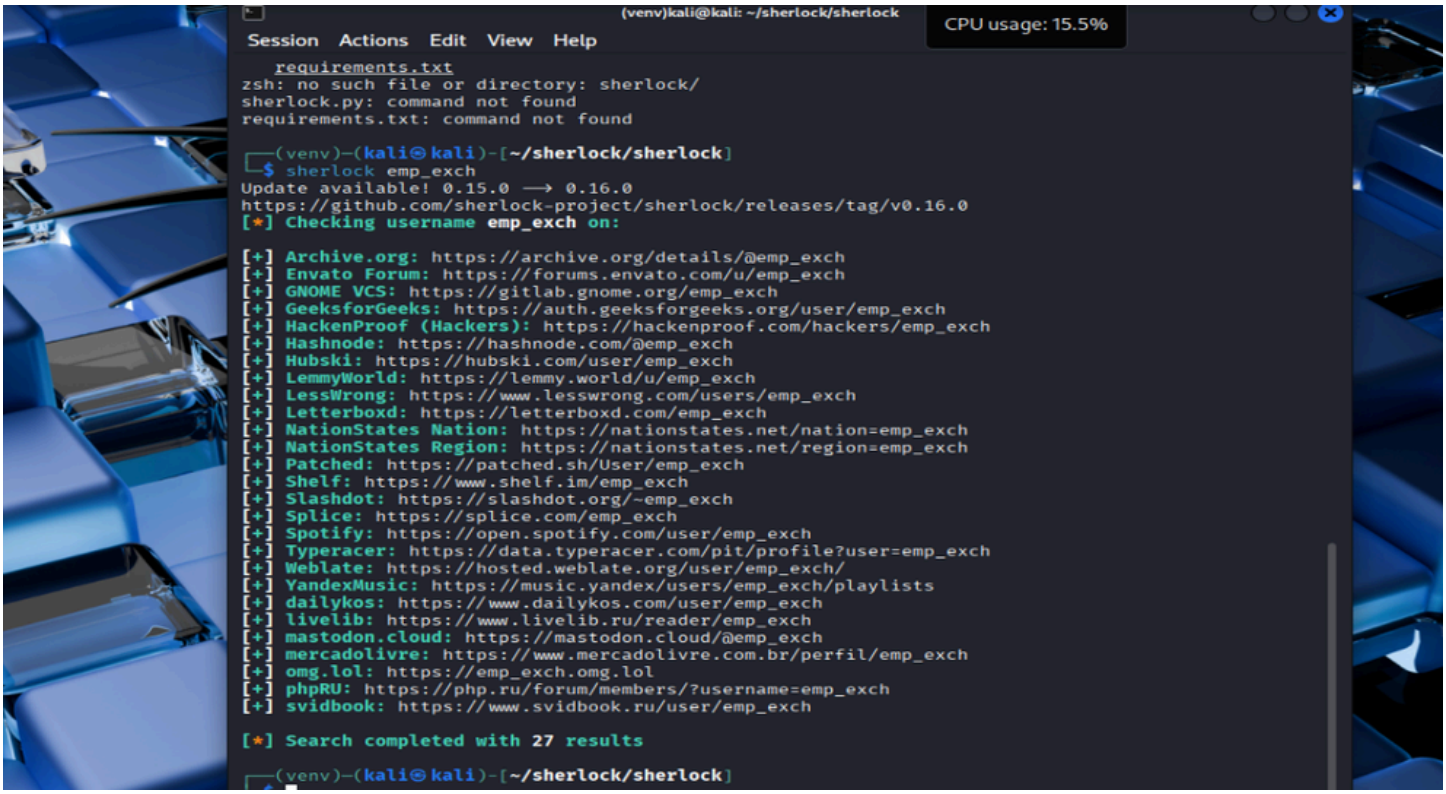
ICQ Found
https://icq.im/emp_exch/en
Social Network | 1.60s

D. Username Enumeration Using Sherlock

> python3 sherlock emp_exch

>sherlock emp_exch --output emp_exch.txt

Findings: Sherlock found multiple matches, but only the X (Twitter) account is active.



E. whoami

Findings: The username **emp_exch** is active on several CTF and tech platforms, making it a useful starting point for further OSINT research.

D. whoami

emp_exch | Categories | STOP

NAME	PREVIEW	CAPTURE	URL	DATA SOURCE	CATEGORY
247ctf	PREVIEW		https://247ctf.com/progress/emp_exch	WhatsMyName	tech
aparat	PREVIEW		https://www.aparat.com/emp_exch/	Sherlock	
archive.org	PREVIEW		https://archive.org/details/@emp_exch	Sherlock	
authorstream	PREVIEW		http://www.authorstream.com/emp_exch/	Sherlock	
avizo	PREVIEW		https://www.avizo.cz/emp_exch/	Sherlock	
bibsonomy	PREVIEW		https://www.bibsonomy.org/user/emp_exch	Maigret	in
bluesky	PREVIEW		https://bsky.app/profile/emp_exch.bsky.social	Sherlock	

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1. **Profile Metadata Collection**

- **Username:** @emp_exch (2 past username changes, last in Oct 2023)
- **Display Name:** emp_exch
- **Bio:** Learning Automation in Online & On-Site MS Exchange.
- **Position:** Associate Technician at PwrGrid Mgmt
- **Location:** Not Specified
- **Joined:** April 2023
- **Followers / Following:** 9 Followers / 16 Following
- **Access Method:** Connected via Web

Findings: Follows major tech & cloud-based accounts such as Microsoft, Azure, AWS, Google, Kubernetes, and IBM Cloud. Interests strongly align with cloud infrastructure & DevOps.

3. **Profile Picture Analysis:**

The profile (in X) shows a simple “G” as the display picture. It’s likely just the first letter of the user’s name or email (Gmail), but this can’t be confirmed.

4. **Cross-Platforms**

A. Email

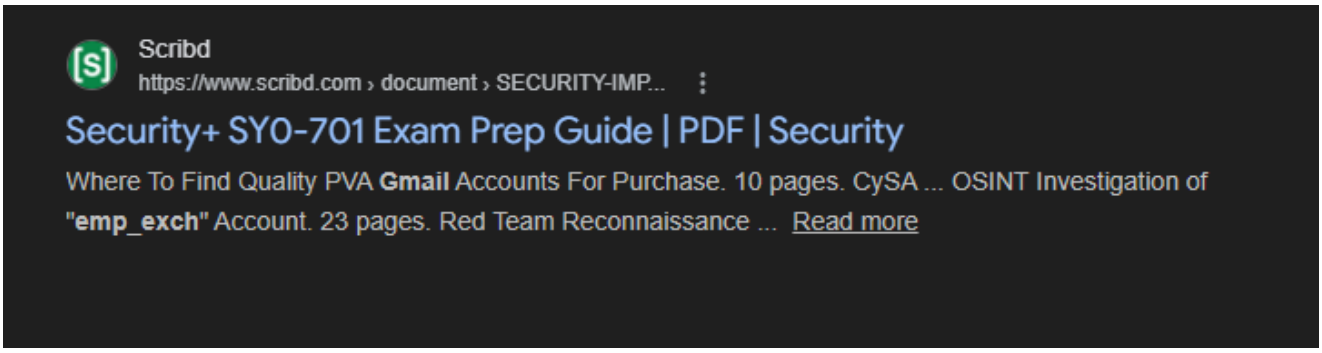
No public email was found.

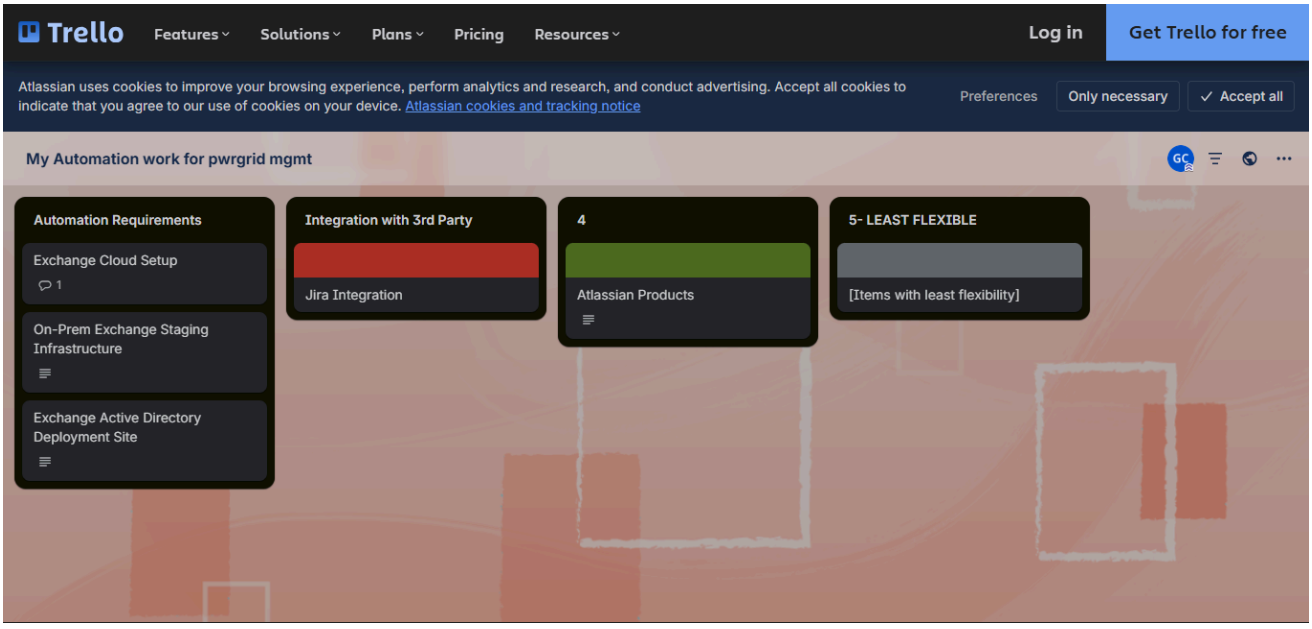
B. WHOIS

- Searched for any domains linked through the identified profiles.
- No domains were found to be registered under the username **emp_exch**.

Correlation Analysis

Some publication and project automation links shared on GitHub were also posted on Twitter, reinforcing the linkage between the accounts.





Final Conclusion

- The username emp_exch appears to belong to someone active in the InfoSec and OSINT community.
- Activity is consistent across platforms.
- Strong DevSecOps observed.
- No clear real-world identity leaks found.

The Vault Report

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2. **Static Analysis**
3. **Disassembly Analysis**
4. **Reconstruction**
5. **Verification**

Scope & Objective

This challenge is a **Linux Binary Reverse Engineering Assessment**

Downloaded the “THE VAULT.zip”.

1. Reconnaissance

File Enumeration

> ls

> file "The Vault.zip"

Out: > The Vault.zip: Zip archive data, made by v2.0,

extract using at least v2.0,

last modified Jan 01 1980 00:00:00,

uncompressed size 15924,

method=deflate

- We confirmed it is a ZIP archive.
- We noticed a suspicious timestamp (1980)
  - The creator intentionally removed metadata
- Not encrypted.

Issue: > file The Vault

Command fails !!!

Findings: Linux interpreted that as:

- The
- Vault

Two separate files

Correct way: file "The Vault.zip"

Archive Inspection

Inspected contents without extracting

> unzip -l "The Vault.zip"

Findings: “The Vault/vault”, it contains a single file. No extension.

Binary Identification

Analysed the file type

> file vault

ELF	Linux executable format
32-bit	Compiled for i386
PIE	Position Independent Executable
Dynamically linked	Uses system libraries
NOT stripped	Debug symbols exist

2. **Strings Analysis**

Objective: Extract readable text from the binary to identify clues like URLs, IPs, or hidden commands.

> strings vault | less

- We split into 4 sections:
    i. Normal System Strings
    ii. Program Logic Strings - Shows how the binary works
    iii. Suspicious Fragments
    iv. Function Names

> Usage: ./vault [OPTIONS]

--flag

--intro

--help

Enter the password to access the Flag:

Better Luck Next Time!

Too many arguments, try again!!!

Findings: The flag is password-protected.

Every string ends with **f.**

The flag might be stored in fragments

There are functions says:

- concatpasswd
- concatflag

Static Analysis with objdump

> objdump -d vault | grep concat

functions are CALLED inside <main>

- Hence, the flag and password are built at runtime.
- We found out the references to the addresses of both functions.
- The flag pieces are passed INTO concatflag from another function, so inspected where concatflag is CALLED.(<main>)

Disassembly Analysis

- **Analyzed main**
- **Reconstructed push order**
- **Decoded little-endian constants**

Analyse the concatflag

Some arguments are pushed in reverse order

144c: lea -0x68(%ebp),%eax → push
1450: lea -0x62(%ebp),%eax → push
1454: lea -0x6e(%ebp),%eax → push
1458: lea -0x50(%ebp),%eax → push

Decode Each Fragment

Findings:

- The flag is stored in memory in hexadecimal fragments using little-endian format.
- Reverse each hex fragment to text and combine them.

Fragment 1: 0x67616c46 + 0x7b → Flag{

Fragment 2: 0x41753059 + 0x72 → Y0uAr

Fragment 3: 0x72434133 + 0x61 → 3ACra

Fragment 4: 0x72336b63 + 0x7d → ck3r}

Reconstructed FLAG

- Built password manually
- Built the flag manually

Flag{Y0uAr3ACrack3r}

Decode Each Fragment of <concatpasswd>

Final Password: S7p3rS3ct3rOrgan1z2t10n

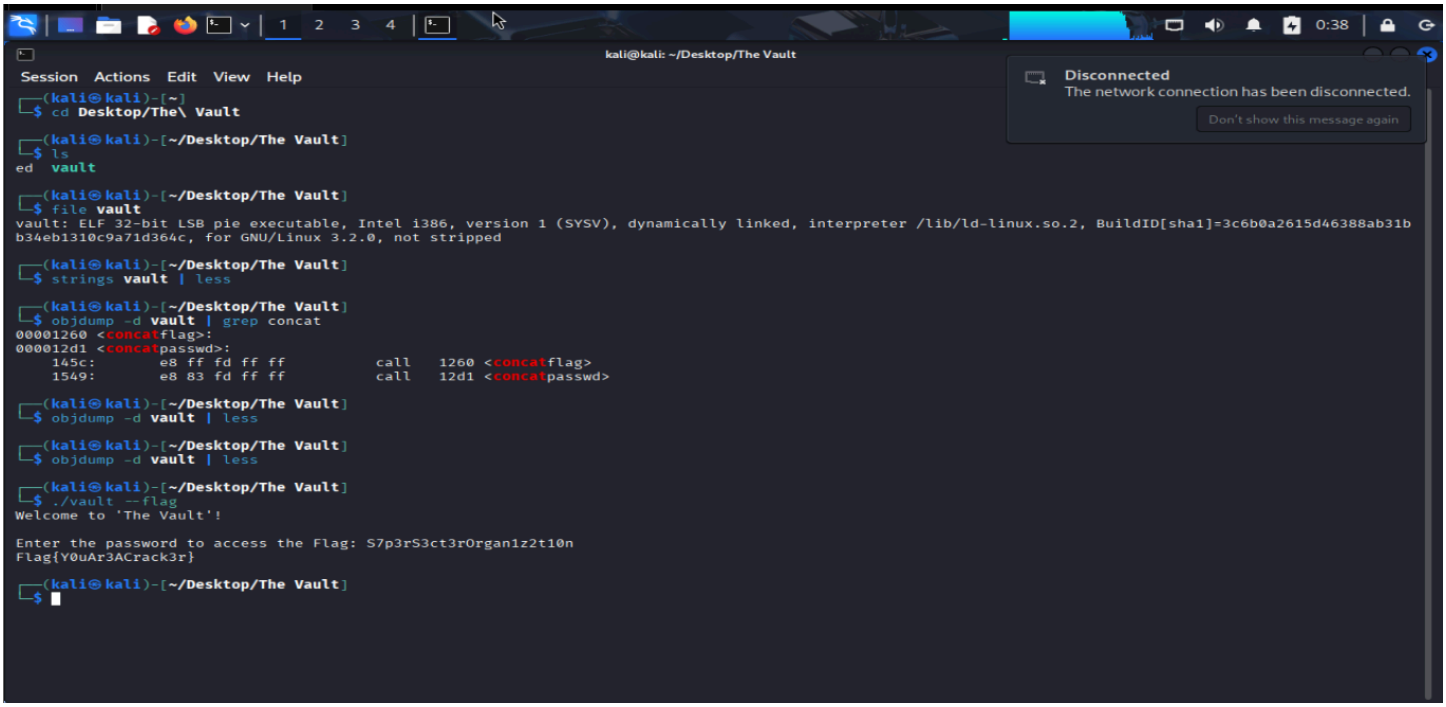
as Final Execution

> ./vault --flag

out: Flag{Y0uAr3ACrack3r}

Conclution

- Reversed stack-based string construction
- Understood calling conventions
- Decoded little-endian hex
- Rebuilt hidden flag
- Rebuilt the hidden password



Forensics Report

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2. **Safe Extraction**
3. **Extract Visible Text**
4. **Check for Hidden URLs**
5. **conclution**

Identify the File

Analysed file name and extension, file command output, and Hex editor info

> ls -la

> file Live_streaming.rar

FIndings:

File Name: Live_streaming.rar
File Size: 289,760 bytes (~283 KB)
Extension: .rar
Real Type: RAR archive data, version 5

File type verified for safe Extraction.

*****Extract a file, especially from an unknown source, there's always a risk of malware*****

1. Precautions Before Extracting

- using an isolated VM or a disposable environment.
- Checked the file hash

  > sha256sum Live_streaming.rar

2. Inspect Before Extracting

- List the contents without extracting:

  > unrar l Live_streaming.rar

3. Scan the RAR

- Scanned with **ClamAV**

> clamscan Live_streaming.rar

Safety Checks:

1. **No suspicious filenames** — contains only a PDF.
2. **ClamAV scan:** The .rar archive contains only one file, " Live_streaming.pdf". The archive itself is clean, no malware detected

Safe Extraction

- Created a separate folder for extraction
- Verified the extracted file
- Scan the PDF again with ClamAV

> unrar x Live_streaming.rar safe_extract/

> ls -la safe_extract/

> file safe_extract/Live_streaming.pdf

> clamscan safe_extract/Live_streaming.pdf

Extraction details:

Extraction Folder: safe_extract/
Extracted File: Live_streaming.pdf
File Size: 422,769 bytes (~412 KB)
File Type: PDF document, version 1.4, 8 pages

Findings: ClamAV scan shows **no viruses detected**

The archive and extracted PDF are clean.

Analyze PDF Metadata

Extracted metadata often contains hidden information (author name, software used, timestamps, etc.).

Tool used: exiftool

> exiftool safe_extract/Live_streaming.pdf

- This PDF is not a random document.
- It is likely a Wikipedia page saved as PDF.
- It was generated using:
  - Headless Chrome (automated browser)
  - Rendering engine: Skia/PDF
- It was created on October 5, 2025.
- The use of HeadlessChrome suggests:
  - It may have been generated programmatically.
  - Possibly scraped or auto-downloaded.

Extract Visible Text

Extracted all readable text from the PDF.

Tool used: pdftotext

> pdftotext safe_extract/Live_streaming.pdf output.txt

Findings: This PDF is a normal, visibly not modified, Wikipedia export.

Note: An employee wouldn't save a normal stream file, hence this means hidden data is present.

Check for Hidden URLs

> strings safe_extract/Live_streaming.pdf | grep -i http

Findings: All links are normal Wikipedia reference links.

- YouTube
- TikTok
- Twitch
- Facebook
- Amazon
- The New York Times
- CNN
- The Guardian

conclution

This PDF is 100% clean. There is no malware, no embedded payload.